Survey of physics faculty insights on the US quantum curriculum & research needs

Alexis Buzzell[1], Ramón Barthelemy[2], & Tim Atherton[3]
[1]Auburn University, [2]University of Utah, [3]Tufts University

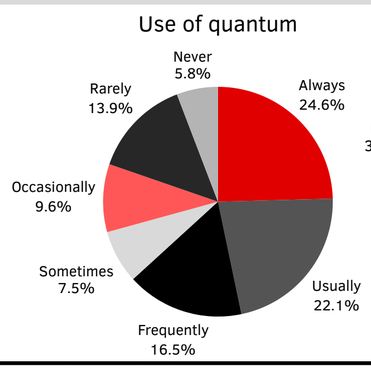
PER U
Physics Education Research at The University of Utah

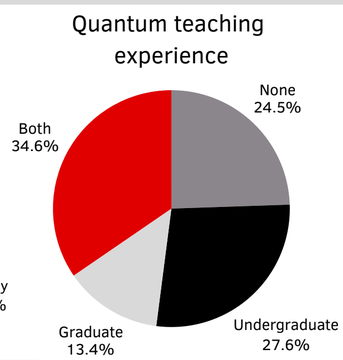
INTRODUCTION & LITERATURE REVIEW

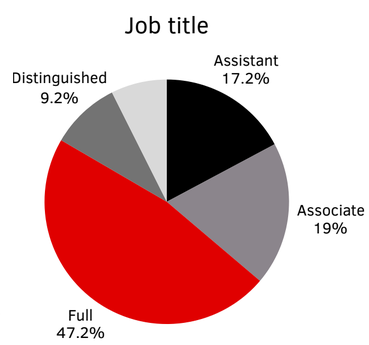
- Progressing from coursework to research is an important milestone in graduate education
  - Typically marked by passing the qualifying exam (QE)
- 90% or more of US physics graduate programs test students' mastery of quantum concepts on the QE [1]
- Undergraduate quantum curriculum analysis found only 34% of institutions taught Dirac notation in required courses [2]
- Sakurai [3] was found to be most common textbook for first semester graduate quantum courses [4]
- PhD students who struggle to find a research group report feelings of isolation and concerns about completing their program in timely manner [5]
  - Those who integrate smoothly have increased sense of belonging and program satisfaction [5]
- Graduate students report feeling underprepared when deciding on which research group to join
- Motivation: Highlight the alignment or misalignment of what is taught and what is applied in contemporary physics research

METHODS

- Survey instrument designed from undergraduate [2] and graduate [4] curriculum analyses
- Employed census-style, non-probability convenience sampling method
  - All 8,304 physics faculty at 194 US institutions were invited to participate
  - n = 551 participated, completed the consent form, and completed the entirety of the survey
- Quantitative analysis:
  - Descriptive statistics to determine demographics
  - $\chi 2$ analysis to determine statistical significance of item response when comparing population groups (faculty use of quantum topics in research, teaching experience, and job title)
- Qualitative analysis:
  - Used for open response question
  - Emergent coding method for thematic analysis



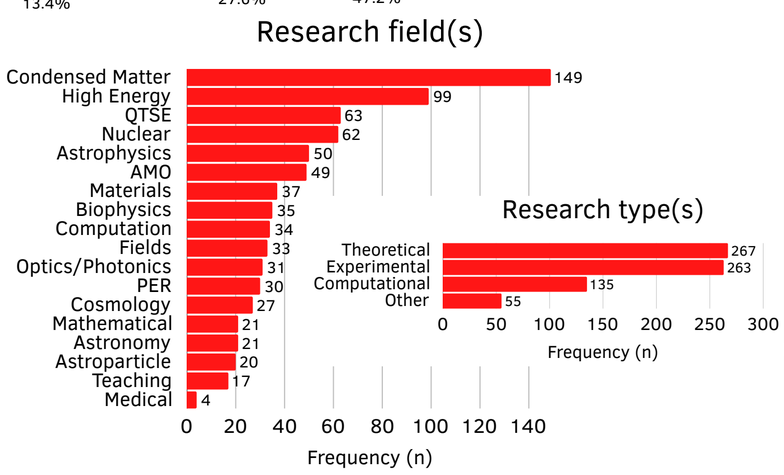




In order for a graduate student in your research group to conduct and complete dissertation-level research, they must understand the following quantum concepts at an expert level.

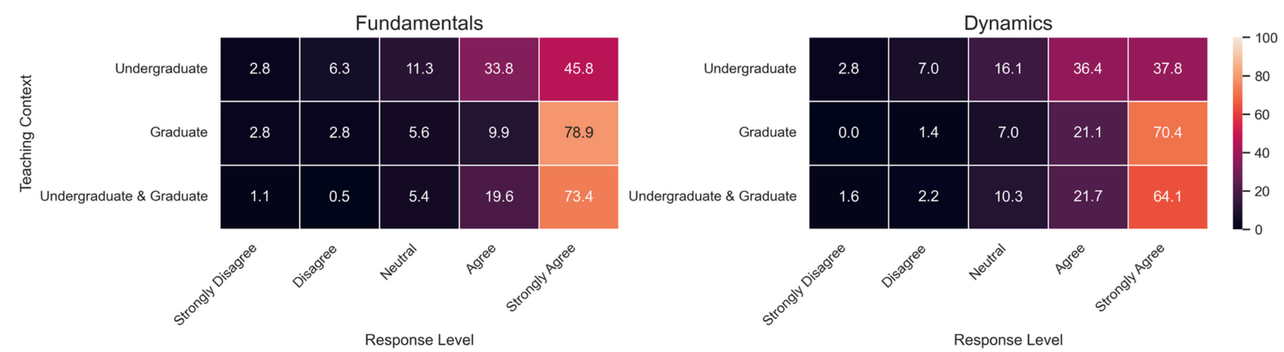
	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Fundamentals (states, observables, operators, Hilbert space, Dirac notation)	○	○	○	○	○
Dynamics (Time evolution, Heisenberg/Schrodinger pictures, Stationary states)	○	○	○	○	○

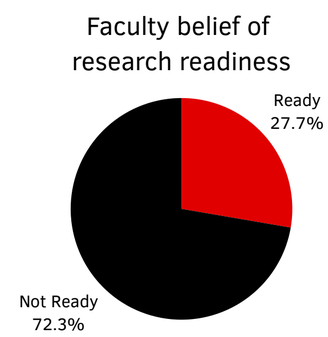
After completion of their coursework, are the students in your research group prepared to conduct dissertation-level research? Are there certain topics or skills that you must provide support on that are not covered adequately, or at all, in the curriculum?

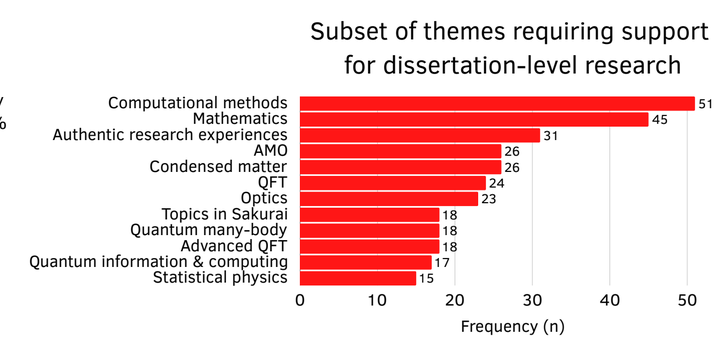


RESULTS

- $\chi 2$ analysis showed those who use quantum more often in research ranked concepts as more important than those who use it less often
- $\chi 2$ analysis showed senior and early-career faculty largely agreed on the importance of concepts
- $\chi 2$ analysis revealed significant differences ($p<0.05$) in responses based on teaching experience across 10 out of the 14 concepts (fundamentals, dynamics, angular momentum, approximations, scattering, identical particles, QFT, ensembles, statistical mechanics, and quantum information)
- Those who have taught graduate courses rank topics with having higher importance than those who have only taught at the undergraduate level
- Majority of faculty (72.3%) do not believe students are prepared for dissertation-level research after completion of coursework
- Most commonly mentioned areas needing support included computational methods, mathematics, and a need for authentic research experiences







DISCUSSION & CONCLUSION

- Differences in teaching experiences are consistent with previous findings that undergraduate coursework does not cover fundamental concepts at the level assumed by graduate courses
  - Faculty who have taught graduate-level quantum expect a different conceptual baseline than faculty who have only taught at the undergraduate-level
- Research readiness is not solely achieved through mastery of formal quantum concepts but through participation in the epistemic practices of a research community
- Improving student preparation for dissertation-level research may therefore include:
  - Integrating research practices earlier
  - Connecting undergraduate and graduate curricula
  - Designing shared core competencies that transcend research fields

[1] Campbell, et. al., 2006. Report of the joint AAPT-APS task force: Graduate education in physics. Technical report, American Association of Physics Teachers and American Physical Society, College Park, MD.
[2] Buzzell, A., et. al., 2025. Quantum mechanics curriculum in the U.S.: Quantifying the instructional time, content taught, and paradigms used. Physical Review Physics Education Research 21(1), 010102.
[3] Sakurai, J.J., Tuan, S.F., 1985. Modern Quantum Mechanics. Benjamin/Cummings Pub., Menlo Park, CA.
[4] Buzzell, A., et. al., 2025. Characterizing the U.S. graduate-level quantum curriculum: Insights into doctoral programs curriculum and comparisons with undergraduate curriculum, PREPRINT (Version 1) available at Research Square [https://doi.org/10.21203/rs.3.rs-7745493/v1]
[5] Verostek, M., et. al., 2024. Physics Ph.D. student perspectives on the importance and difficulty of finding a research group. Physical Review Physics Education Research 20(1), 010136.